

COTS Hardware Nomenclature

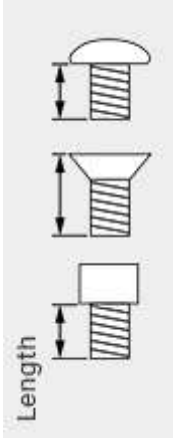
Metric Screws

P/N - M#x## HEAD *Details

M Thread x Length

Head:

- Button Head (BHD)
- Socket Head (SHD)
- Countersink (CSK)
- Low Profile Socket Head (LPSH)
- Extra Low Profile Socket Head (XLPSH)
- Hex Head (HEX)
- Flanged Hex Head (FHX)
- Serrated Flange Hex Head (SFHX)
- Flanged Button Head (FBH)
- Ultra Wide Flanged Button Head (UWFBH)
- Extra Wide Button Head (XWBH)



Hex vs torx

Details (Optional):

- Material
- Surface Treatment
- Drive
- Thread Length
- Thread Direction
- Etc.

Prioritize McMaster Carr hardware
Prioritize metric hardware
Prioritize black-oxide hardware

Ex.

M6x1 25mm Long Flanged Button Head Screw
Alloy Steel Black-Oxide

- 92137A540 - M6x25 FBH

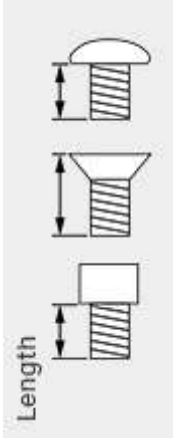
English Screws

P/N - #-##x## HEAD *Details

Thread x Length

Head:

- Button Head (BHD)
- Socket Head (SHD)
- Countersink (CSK)
- Low Profile Socket Head (LPSH)
- Extra Low Profile Socket Head (XLPSH)
- Hex Head (HEX)
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- Flanged Button Head (FBH)
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- Extra Wide Button Head (XWBH)



Hex vs torx

Details (Optional):

- Material
- Surface Treatment
- Drive
- Thread Length
- Thread Direction
- Etc.

Prioritize McMaster Carr hardware
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Prioritize black-oxide hardware

Ex.

1/4"-20 1" Long Hex Head Grade 5 Screw Black-Oxide

- 92965A542 - 1/4-20x1 HEX

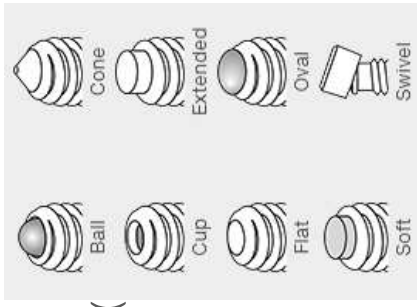
Metric Set Screws

P/N - M#x##TIP SET *Details

M Thread x Length

Tips:

- Cup Point (CP)
- Knurled Cup Point
- Flat Point (FL)
- Soft Point (SF)
- Oval Point (OV)
- Extended Point
- Cone Point (CN)



(

(EX)

Details (Optional):

- Material
- Vent
- Drive
- Etc.

Prioritize using McMaster Carr hardware

Prioritize using metric Hardware

Ex.

M4x0.7 Knurl-Grip Cup-Point 6mm Long Set

Screw Alloy Steel

- 94285A360 - M4x6KC SET

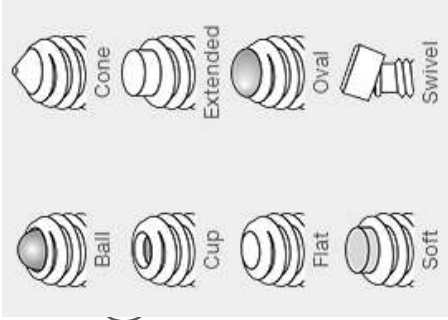
English Set Screws

P/N - #-##x##TIP SET *Details

Thread x Length

Tips:

- Cup Point (CP)
- Knurled Cup Point
- Flat Point (FL)
- Soft Point (SF)
- Oval Point (OV)
- Extended Point (EX)
- Cone Point (CN)



Details (Optional):

- Material
- Vent
- Drive
- Etc.

Prioritize McMaster Carr hardware
Prioritize metric Hardware

Ex.

5-40 Cone-Point 1/4" Long Set Screw Black-Oxide Alloy Steel

- 92695A125 - 5-40x1/4CN SET

Shoulder Screws

P/N - M#x##-D#x## HEAD *Details

M Tread x Length - D Shoulder x Length

Head:

- Socket Head (SHD)
- Low Profile Socket Head (LPSH)
- Ultra Low Profile Socket Head (ULPSH)
- Countersink (CSK)

Details (Optional):

- Material
- Tolerance
- Etc.

Prioritize metric hardware

Ex.

6mm Shoulder Diameter, 20mm Shoulder Length, M6 Shoulder Bolt Alloy Steel

- 90782A115 - M6x10-D6x20 SHD

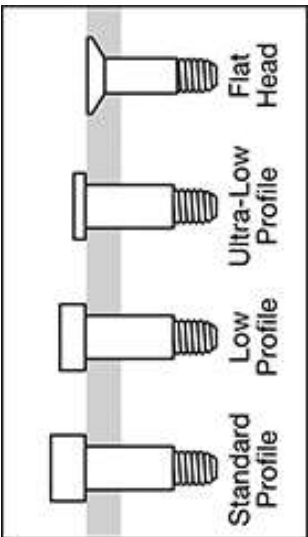
5.5mm Shoulder Diameter, 14mm Shoulder Length, M4 7mm Thread h7 4137 Alloy Steel

- HDMSB5.5-14-F7-M4 - M4x7-D5.5x14 SHD h7

12mm Shoulder Diameter, 8mm Shoulder

Length, M8 8mm Thread 304 Stainless Steel

- DBGSY8-8-8 - M8x8-D12x8 ULPSH



Stand-offs(Coupling Nuts)

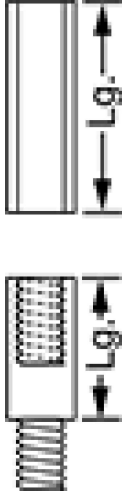
P/N - M#x#F-#x##-M#x#T STANDOFF *Details

M Thread x Thread Length F- Xsection x Length
- M Thread x Thread Length M/F

Details (Optional):

- Profile (Hex vs. Round)
- Material
- Thread Direction
- Etc.

Prioritize McMaster Carr hardware
Prioritize metric hardware



Ex.

M4x9.5 Male to M4x11.1 Female 19mm Long 8mm Cross-Section Hex Stand-off Aluminum

- 98952A152 - M4x11.1F-8x19-M4x9.5M STANDOFF

M3 THRU 11mm Long 6mm Cross-Section Hex Stand-off 18-8 Stainless Steel

- 94868A738 - M3x5.5F-6x11-M3x5.5F STANDOFF

M8x12.7 Male to M6x11.1 Female 19.1mm Long 13mm Cross-Section Hex Stand-off 18-8 Stainless Steel

- 92499A631 - M6x11.1F-13x19.1-M8x12.7M STANDOFF

M3.5x9.5 Female to M3x6.4 Female 37mm Long 6mm Cross-Section Hex Stand-off Zinc Plated Steel

- 92912A103 - M3.5x9.5F-6x37-M3x6.4F STANDOFF

McMaster Carr Brass Heat Inserts

P/N - M#S/L(T/F) HEATSET		Ex.	
M Thread Short/Long Taper/Flange Prioritize metric hardware		Standard	
		Flanged	
		Tapered	
		M3x0.5 3.8mm Long Tapered Heat Insert	
		● 94180A331 - M3ST HEATSET	
		M6x1.0 12.7mm Long Tapered Heat Insert	
		● 94180A373 - M6LT HEATSET	
		M4x0.7 4.7mm Long Heat Insert	
		● 94459A150 - M4S HEATSET	
		M6x1.0 8.89mm Long Flanged Heat Insert	
		● 97171A360 - M6SF HEATSET	

Dowel Pins

P/N - D#x## FIT PIN *Details

D Diameter x Length

Details (Optional):

- Material
- Vent
- Groove/Knurling
- Surface Treatment
- Tapped
- Etc.

Prioritize McMaster Carr hardware

Prioritize metric hardware

Ex.

5mm Diameter 16mm Long Dowel Pin 52100 Alloy Steel +0.004/+0.012

- 91595A352 - D5x16 m6 PIN

3mm Diameter 10mm Long Undersized Dowel Pin 12L14 Carbon Steel -0.014/+0.000

- 97049A207 - D3x10 h8 PIN

Spring Pins

P/N - D#x## ROLL TYPE *Details

(D Diameter x Length)

Types:

- Slotted (SL)
- Coiled (CL)
- Vibration Resistant (VR)

Details (Optional):

- Material
- Wall Thickness
- Etc.

*Prioritize McMaster

Slotted Spring Pins



Also known as roll, tension, split, and expansion pins, these pins have a slot along one side that you squeeze closed to install them into unthreaded holes.



Vibration-Resistant Slotted Spring Pins

Interlocking teeth help these pins resist sliding when exposed to vibration. They have a slot that presses closed when the pins are inserted so they don't flex after installation.



Coiled Spring Pins

These spiral pins remain flexible after installation, so they absorb shock and vibration better than slotted spring pins. They work well in holes that are out of round.

Ex:

8mm Diameter, 22mm Long Slotted Spring Pin
18-8 Stainless Steel

- 91610A805 - D8x22 ROLL SL

4mm Diameter, 14mm Long Heavy Duty Coiled
Spring Pin 1070-1095 Carbon Steel

- 95755A162 - D4x14 ROLL CL

6mm Diameter, 10mm Long Vibration Resistant
Spring Pin

- SPP-6-10 - D6x10 ROLL VR

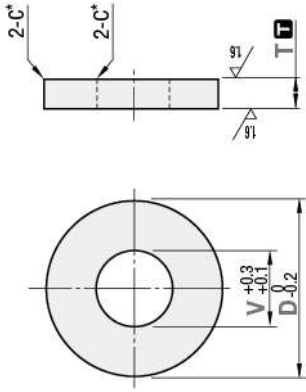
Misumi Circular Spacers

P/N - #x#x# SPACER *Details

ID x OD x Length

Details (Optional):

- Material
- Surface Treatment
- Tolerance
- Etc.



Ex.

10mm ID, 16mm OD, 1mm Thick Spacer

- MSRB10-1.0 - 10x16x1 SPACER

6mm ID, 14mm OD, 2.5mm Thick Spacer 1045 Carbon Steel Electroless Nickel Plating

- WSSM14-6-2.5 - 6x14x2.5 SPACER

9mm ID, 15mm OD, 8mm Thick Spacer 2000 Series Aluminum Black Anodized $+0/-0.2$ L Tol.

- AWSAAB-D-D15-V9-T8 - 9x15x8 SPACER

7mm ID, 17mm OD, 34mm Length Spacer, 1045 Carbon Steel, Low Temp Black Chrome Plate, g6 Outer Tol, H7 Inner Tol, $+0/-0.1$ Length Tol.

- SNCR-GH-V7-D17-L34 - 7x17x34 SPACER

Circular Shims

P/N - #x#x# SHIM *Details

ID x OD x Length

Details (Optional):

- Material
- Tolerance
- Etc.

Suppliers:

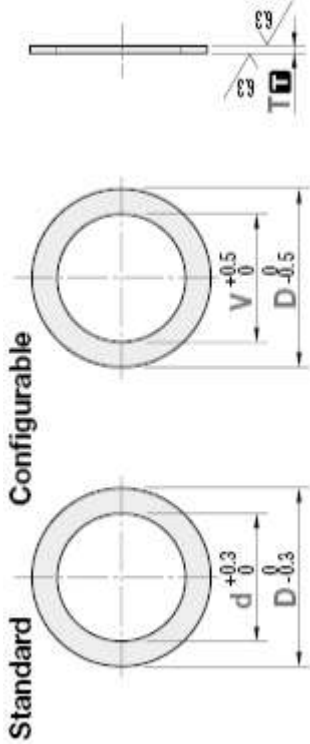
- McMaster Carr
- Misumi
- Shim Shack

10mm ID, 18mm OD, 0.05mm Thick Shim Ring
304H Stainless Steel

- PCIMRS10-18-0.05 - 10x18x0.05 SHIM

16mm ID, 22mm OD, 0.30mm Thick Shim Ring
1074-1095 Spring Steel

- 98055A221 - 16x22x0.3 SHIM



Metric Round O-Rings

ORING #.##X# Material *Details

Xsection x Inner Diameter

Details (Optional):

- Supplier
- Supplier Part Number
- Color
- Etc.

Ex.

5mm ID, 9mm OD, 2mm Thick 70A Durometer
Buna-N O-Ring

- ORING 2.00x9 Buna-N 70

Washers

P/N - #x#x# TYPE WASH *Details

ID x OD x Thickness

Type:

- | | |
|------------------|-------|
| • General | () |
| • Countersunk | (CSK) |
| • Finishing | (FNH) |
| • Clipped | (CLP) |
| • Double Clipped | (DCP) |
| • Spherical | (SPH) |

Details (Optional):

- Material
- Surface Treatment
- Tolerance
- Etc.

Prioritize McMaster Carr hardware
Prioritize metric hardware

Ex.

- M5 General Purpose 18-8 Stainless Steel Washer 9mm OD
- 98689A114 - 5.3x9x1 WASH
- M3 Finishing Countersunk Washer Black-Oxide Steel 90deg Countersink Angle 12mm OD
- 92908A576 - 3.2x12x2.5 FNH WASH
- M8 Spherical Washer 3deg Misalignment 17mm OD Steel 5.1-5.8mm Thick
- 98148A154 - 8.4x17x5.45 SPH WASH

Lock Washers

P/N - #x#x# TYPE LWASH *Details

ID x OD x Thickness

Type:

- Split Lock (SL)
- External Tooth Lock (XT)
- Internal Tooth Lock (NT)
- Countersunk External Tooth Lock (CKXT)
- Belleville Lock (BL)
- Serrated Belleville Lock (SRI)

Details (Optional):

- Material
- Etc.

Prioritize McMaster Carr hardware
Prioritize metric Hardware

Ex.

- M7 External Tooth Lock Washer 12.5mm OD
18-8 Stainless Steel
- 95060A460 - 7.4x12.5x0.8 XT LWASH
- M5 Belleville Spring Washer 11mm OD 17-7 PH
Stainless Steel
- 91235A316 - 5.3x11x0.9 BL LWASH

Nuts

P/N - M# TYPE NUT *Details

Type:

- Standard ()
- Serrated (SER)
- Flanged (FLG)
- Serrated Flange (SRFL)
- Socket (SOC)

Details (Optional):

- Class/Grade
- Material
- Surface Treatment
- Thread Direction
- Thickness
- Etc.

Prioritize McMaster Carr hardware

Prioritize metric hardware

Ex.

M12x1.75 Brass Thin Hex Nut

- 93187A400 - M12 NUT Brass Thin

M5x0.8 Serrated Flange Nut Class 8 Steel

Black-Oxide

- 94920A120 - M5 SRFL NUT

M3x0.5 Alloy Steel Socket Nut

- 92066A113 - M3 SOC NUT

Lock Nuts

P/N - M# TYPE *Details

Type:

- Nylock (NYLOCK)
- Keps (KEPS NUT)
- Nonmarring Keps (NM KEPS NUT)
- Distorted Thread (DIST THR NUT)
- Flex Top (FLEX NUT)
- Spring Stop (SPRING NUT)
- Flange Nylock (FL NYLOCK)
- Flange Distorted Thread (FL DIST THR NUT)

Details (Optional):

- Class/Grade
- Material
- Surface Treatment
- Thread Direction
- Thickness
- Etc.

Prioritize McMaster Carr hardware

Prioritize metric hardware

Ex.

M8x1.25 Nylock Nut 18-8 Stainless Steel

- 93625A300 - M8 NYLOCK

M5x0.6 Spring-Stop Nut Zinc-Plated Class 6 Steel

- 90508A110 - M5 SPRING NUT

M3x0.5 Nonmarring Keps Nut Zinc-Plated Steel Class 8

- 91270A250 - M3 NM KEPS NUT

PEM

P/N - M#x## TYPE PEM *Details

M Thread x Length

Type:

- Round Nut (RND)
- Hex Nut (HEX)
- Flush Nut (FLH)
- Aligning Nut (ALN)
- Nylock Nut (NYL)
- Stand off (SDF)
- Pin (PIN)
 - D#x## PIN PEM
- Stud (STD)

Details (Optional):

- Material
- Etc.

Prioritize McMaster Carr hardware
Prioritize metric hardware

Ex.

M3x0.5 1.5mm Thick Press-Fit Nut Zinc-Plated Steel

- 95185A530 - M3x1.5 RND PEM

M5x0.8 20mm Long Press-Fit Stud Zinc-Plated Steel

- 93580A606 - M5x20 STD PEM

1.5mm Diameter 6mm Long Captive Pin A286 Stainless Steel

- 95648A280 - D1.5x6 PIN PEM

Retaining Rings

P/N - ## DIR TYPE RING *Details

Diameter INT/EXT

Types:

- Standard (SNAP)
- Push-On (PUSH)
- E-Clip (E)
- Single Turn (SNGT)
- Spiral (SPRL)
- Inverted (INV)

Details (Optional):

- Material
- Surface Treatment
- Heavy Duty
- Bowed
- Chamfered
- Etc.

Prioritize McMaster Carr hardware
Prioritize metric hardware

Ex.

30mm Spiral External Retaining Ring, 1.1mm Thick,
1060-1090 Spring Steel

- 92623A314 - 30mm EXT SPRL RING

10-14mm Side-Mount External Retaining Ring,
1.1mm Thick, Black-Phosphate 1060-1090 Spring
Steel

- 98543A117 - 10-14mm EXT E RING

20mm Inverted Internal Retaining Ring, 1mm Thick,
Black-Phosphate 1060-1090 Spring Steel

- 90442A208 - 20mm INT INV RING

Bushings

Details: material, manufacturer

P/N - #x#x# TYPE BUSHING *Details

ID x OD x Length (not including flange)

Type:

S - Straight

##F - Diameter Flange

Thrust bushing

Bearings

P/N - #x#x# TYPE BEARING *Details

ID x OD x Length

- Flanged vs Standard Thrust
- Angular contact Combination
- Tapered Roller Needle
- Crossed Roller
- Spherical Roller Roller

Misumi Shafts

P/N - D#x## SHAFT

<https://us.misumi-ec.com/vona2/detail/110300007460/?searchFlow=results2products&KWSearch=configurable%20shafts&list=PageSearchResult>

<https://us.misumi-ec.com/vona2/detail/110300007370/?searchFlow=results2products&KWSearch=configurable%20shafts&list=PageSearchResult>

Misumi Hinge Pins

P/N -

Clevis pin/made to order

<https://us.misumi-ec.com/vona2/detail/110300098370/?searchFlow=results2products&KWSearch=HINGE%20pin&list=PageCategory>